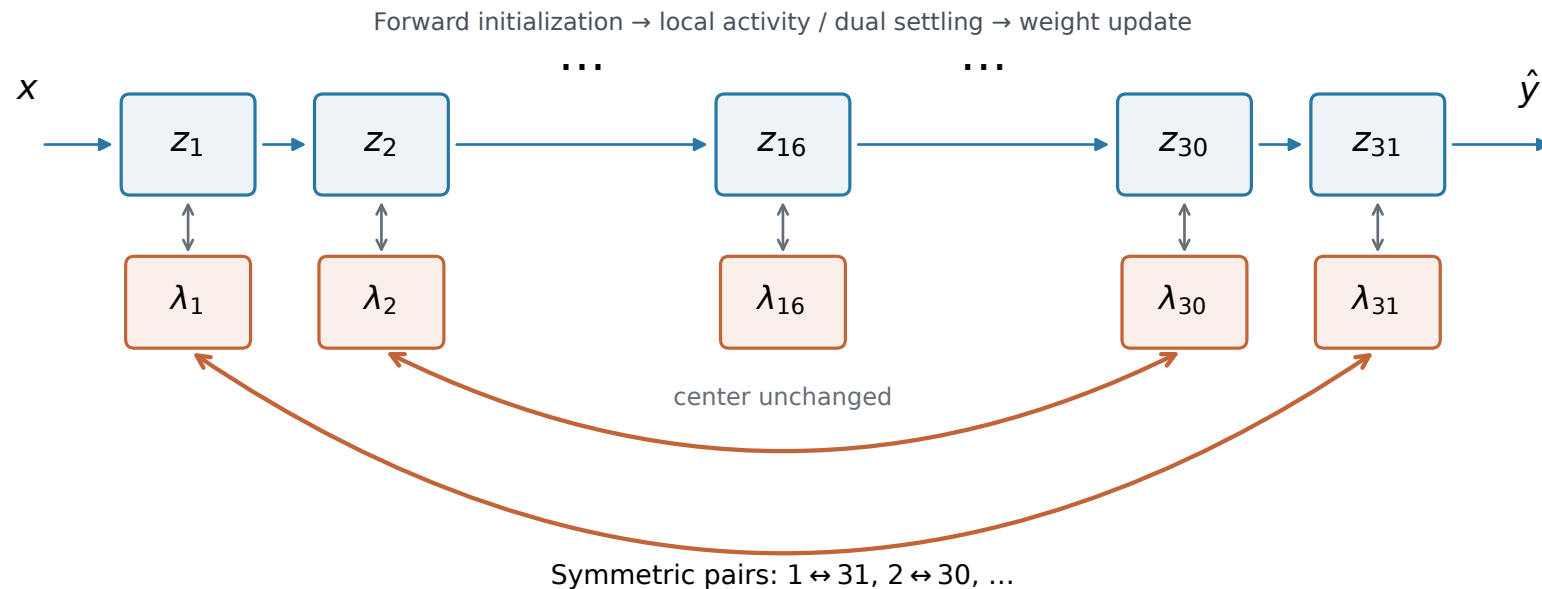


Elastic Dual Coupling

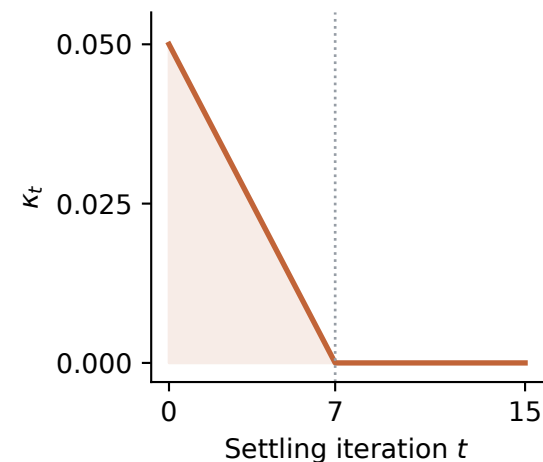
(a) Separate multipliers, softly connected across depth



$$\lambda_i^{t+1} = \lambda_i^t + \alpha r_i(z^{t+1}) - \kappa_t (\lambda_i^t - \lambda_{H+1-i}^t)$$

$$\kappa_t = 0.05 \max(1 - t/(\lfloor T/2 \rfloor - 1), 0), \quad \alpha = 1, H = 31$$

(b) Fade, then refine independently



Example: $T = 16$
Zero coupling from $t = 7$ onward

GDI retrieves the initial multipliers from training memory. Both partners read old duals; no extra gradient evaluations.